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- 1 Basic queries (SELECT, WHERE)
- 2 Joins (INNER, LEFT)
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7. 🧠 NEW SECTION: AI FOR TESTING

Topics you must cover:

1. What is Prompt Engineering
2. How AI generates test cases
3. AI vs deterministic testing
4. What is agentic workflow
5. How to test AI responses

CORE CONCEPTS

PLAYWRIGHT → CORE CONCEPTS (COMPLETE)

We will cover:

1. Architecture
 2. Installation & Project Setup
 3. Test Runner (test, expect)
 4. Browser Context & Page
-

1 PLAYWRIGHT ARCHITECTURE

1 What is it

Playwright architecture defines how tests interact with browsers using a **Node.js layer + browser engines (Chromium, Firefox, WebKit)**.

2 Key Components

- Test Script (JS/TS)
 - Playwright API
 - Browser Server
 - Browser Engines
 - Browser Context
 - Page
-

3 How YOU should answer (Interview Style)

Playwright follows a client-server architecture where test scripts communicate with browser engines through Playwright APIs. It supports multiple browsers like Chromium, Firefox, and WebKit, and uses browser contexts for isolated test execution.

4 Practical Example

```
</> JavaScript   
  
test('example', async ({ page }) => {  
  await page.goto('https://example.com');  
});
```

👉 Script → Playwright → Browser → Execution

5 Common Mistakes

- ✗ Saying “Playwright only works on Chrome”
 - ✗ Not mentioning **browser context**
 - ✗ Over-explaining internal working
-

6 Expected Questions

- What is Playwright architecture?
 - How is it different from Selenium?
 - What is browser context?
-

2 INSTALLATION & PROJECT SETUP

1 What is it

Setting up Playwright project using Node.js and installing required dependencies.

2 Key Components

- Node.js
 - npm
 - @playwright/test
 - Browsers installation
 - Project structure
-

3 How YOU should answer

I set up Playwright using Node.js by installing @playwright/test via npm. Then I install browser binaries and create a test folder structure for writing test cases.

4 Practical Example

```
</> Bash
npm init -y
npm install -D @playwright/test
npx playwright install
```

5 Common Mistakes

- ✗ Saying "Playwright needs Selenium"
- ✗ Not knowing commands
- ✗ Not knowing project structure

6 Expected Questions

- How do you install Playwright?
 - What are prerequisites?
 - What is project structure?
-

3 TEST RUNNER (test, expect)

1 What is it

Playwright Test Runner is used to execute tests and perform assertions.

2 Key Components

- `test()`
 - `expect()`
 - hooks (`beforeEach`, `afterEach`)
 - fixtures
-

3 How YOU should answer

Playwright provides a built-in test runner where `test()` defines test cases and `expect()` is used for assertions. It supports fixtures and hooks for better test management.

4 Practical Example

```
JavaScript

test('title check', async ({ page }) => {
  await page.goto('https://example.com');
  await expect(page).toHaveTitle(/Example/);
});
```

5 Common Mistakes

- ✗ Confusing test() with function
- ✗ Not knowing expect() usage
- ✗ Ignoring hooks

6 Expected Questions

- What is test() in Playwright?
- What is expect()?
- What are hooks?

4 BROWSER CONTEXT & PAGE

1 What is it

Browser context is an isolated session, and page represents a tab.

2 Key Components

- Browser
- Context
- Page

3 How YOU should answer

Browser context is like a separate session similar to incognito mode, and page represents a single tab. This helps in running tests independently without affecting each other.

4 Practical Example

```
</> JavaScript  
  
const context = await browser.newContext();  
const page = await context.newPage();
```

5 Common Mistakes

- ✗ Saying context = browser
 - ✗ Not understanding isolation
 - ✗ Confusing page and context
-

6 Expected Questions

- What is browser context?
- Difference between browser and context?
- Why is context important?

LOCATORS



PLAYWRIGHT → LOCATORS (PART 1)

We'll cover in parts:

1. locator()
 2. getByRole() ★
 3. getByText()
 4. CSS & XPath
 5. Best Practices
-



1 locator()

1 What is it

`locator()` is a Playwright method used to find elements using selectors like CSS, XPath, id, etc.

2 Key Components

- CSS selectors (`#id`, `.class`)
 - XPath
 - Chaining (`locator().locator()`)
 - Auto-waiting support
-

3 How YOU should answer (Interview Style)

`locator()` is a flexible method in Playwright used to identify elements using CSS or XPath selectors. It supports auto-waiting and allows chaining for better element targeting.

4 Practical Example

JavaScript

```
await page.locator('#username').fill('Kalpesh');  
await page.locator('.login-btn').click();
```

5 Common Mistakes

- ✗ Using long XPath unnecessarily
- ✗ Not using chaining
- ✗ Over-relying on CSS when better options exist

6 Expected Interview Questions

- What is locator() in Playwright?
- Difference between locator() and getByRole()?
- Why is locator better than findElement (Selenium)?

2 `getByRole()` **(MOST IMPORTANT)**

1 What is it

Used to locate elements based on **ARIA roles (accessibility-based)**.

2 Key Components

- Role (button, textbox, link)
 - Name (visible label/text)
 - Accessibility tree
-

3 How YOU should answer

`getByRole()` is the most recommended locator in Playwright as it uses accessibility roles, making tests more stable, readable, and aligned with real user interactions.

4 Practical Example

 JavaScript

```
await page.getByRole('button', { name: 'Login' }).click();
```



5 Common Mistakes

- ✗ Not using role-based locators
 - ✗ Ignoring accessibility
 - ✗ Using XPath instead
-

6 Expected Interview Questions

- Why is getByRole preferred?
 - What is the ARIA role?
 - Difference between getByRole and locator?
-

3 getText()

1 What is it

Used to locate elements using **visible text on the page**.

2 Key Components

- Exact text match
 - Partial text match
 - Case sensitivity
 - Works on visible UI text
-

3 How YOU should answer (Interview Style)

getText() is used to locate elements based on visible text. It is simple and readable but less stable compared to role-based locators since UI text can change frequently.

4 Practical Example

<> JavaScript



```
await page.getText('Login').click();  
await page.getText('Submit').click();
```

5 Common Mistakes

- ✗ Overusing text-based locators
- ✗ Not handling dynamic text
- ✗ Using when role-based locator is better

6 Expected Interview Questions

- When do you use `getText()`?
 - What are limitations of text locators?
-

4 CSS & XPath

1 What is it

Traditional selector strategies used to locate elements in DOM.

2 Key Components

- CSS (`#id`, `.class`, `tag`)
 - XPath (`//`, `contains()`, `text()`)
-

3 How YOU should answer (Interview Style)

CSS and XPath are fallback locator strategies. CSS is generally faster and preferred, while XPath is useful for complex DOM traversal when other locators are not sufficient.

4 Practical Example

 JavaScript

```
await page.locator('#username').fill('test'); // CSS
await page.locator('//button[text()='Login"]').click(); // XPath
```

5 Common Mistakes

- ✗ Writing long and complex XPath
 - ✗ Using XPath when better locators exist
 - ✗ Not optimizing selectors
-

6 Expected Interview Questions

- CSS vs XPath?
 - Which is faster and why?
 - When do you use XPath?
-

5 Best Practices (INTERVIEW GOLD)

1 What is it

Guidelines to write **stable, maintainable locators**.

2 Key Components

- Prefer getByRole()
 - Use data-testid
 - Avoid XPath
 - Keep selectors simple
 - Use chaining
-

3 How YOU should answer (Interview Style)

My approach is to prefer role-based locators like getByRole for stability. If not available, I use data-testid or CSS selectors. I avoid XPath unless necessary and focus on maintainable and readable locators.

4 Practical Example

</> JavaScript



```
await page.getByRole('button', { name: 'Submit' }).click();  
await page.locator('[data-testid="login-btn"]').click();
```

5 Common Mistakes

- ✗ Using absolute XPath
- ✗ Writing brittle selectors
- ✗ Ignoring readability

6 Expected Interview Questions

- What is your locator strategy?
- Which locator do you prefer and why?
- How do you write stable locators?